How Does Antivirus Work: Detecting Viruses 101

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If you use your computer regularly, which most people do now, then you likely acknowledge the need for antivirus software. There are quite literally millions of viruses and other threats that could make their way onto your computer through downloads, browsing, and other typical internet use. What's worse is that if you bank online or do any shopping, your credit card and bank information are susceptible to hackers.

An antivirus program helps to prevent any of them from accessing your system and data. However, even though you might have antivirus software installed on your device and you know it is always running behind the scenes to keep you protected, learning how does antivirus work is essential knowledge. That said, if you learn how it works a bit better, you are more equipped to keep the software updated and ready to guard against viruses and other threats.

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How Does Antivirus Work to Detect Viruses

In layman's term, you could think of antivirus software as an abstract system working silently to help keep you protected. To understand it better, let us look into viruses and antiviruses in more detail.

Viruses

Viruses, though they sound complicated, are actually quite simplistic in their design. They are just applications that run on a computer, and anyone who knows any advanced programming can alter or produce a new virus.

With that in mind, they are very dangerous, and their simplistic nature means they are more easily created and distributed. Viruses and regular software differ very little in design and application. In fact, it is only their purpose that separates them.

Viruses are only different from regular software applications because they are intended to be harmful, usually by deleting your data, taking your personal information, or crashing your

computer. Similar to other applications, viruses are just a compilation of bits; the sequence of which is called the signature.

Antivirus at Work

A virus leaves virtual fingerprints by using its unique signature while it moves from one system to another. Given that the virus uses this same signature everywhere, the software can remember the signature to discover it if it ever makes its way into your system.

What makes viruses harder to track are the variations that people create to add on to an existing virus signature. For example, the same virus might have a number of different names. That is because people take virus codes that already exist, and they change it to their specifications.

Then they redistribute it. Given that much of the code is the same as the original, the new variation has a similar signature that has only slightly been altered.

Virus Signatures

The virus signature is how antivirus software defends your device from malware. The software program uses a database of virus signatures and checks files for those signatures.

For example, perhaps you decided to click on a link with which you weren't entirely familiar. That link may be a virus disguised as something else, such as a useful program.

Your antivirus software checks the file against its database and realizes the signature is that of a virus. In the above example, where only a slight variation was made to a known virus, the signature may be just different enough to evade the antivirus software's database.

New viruses are created at an alarming rate, and they are more established every single day. This is often why you'll hear computer people mention that you should update your definition files.

This simply means you need to add all of these new viruses to your system's database of virus signatures. If you don't update your database, your software will not know the new virus signature and won't be able to identify it as a dangerous virus.

Keeping the database up to date is truly your first defense against viruses making their way onto your device.

Possible Disadvantages of Using Antivirus Software

There are very few disadvantages to using antivirus software. Any that do exist are usually worth it, given how important the software is in protecting your data and information.

One disadvantage is that many software programs will tend to slow down your overall computer performance because this is precisely how the software works. It has to check all of the executable files on your system against the database of viruses. This is a lot of work being done in the background, and subsequently, the work can slow down your system.

You might not see the work being done, but it is taking place. All you will see is that the system seems to be running slower than usual.

Fortunately, you can temporarily combat this disadvantage. If you know that you need to perform a function quickly or run software fast, you can momentarily disable your antivirus software. A perfect example of when this might be a good idea is if you enjoy playing games on your computer.

Having a slow system is not conducive with gaming. So, if you want to play your game at its normal speed, you'll likely need to disable the antivirus software while playing.

There are other ways to mitigate a slow system over the longer term as well. To start, you can make sure background checks are turned off. You can also schedule the software to run when you are least likely to be using your computer, such as during overnight hours.

That said, you still want to make sure you have your software configured to run enough to protect your system from viruses and other internet threats. Also, if you momentarily disable it to play games or perform some other function that requires fast speed, you want to make sure you turn it back on when you are finished.

Conclusion

Antivirus software is an absolute must in keeping your information protected. Knowing how it works means that you are better equipped to keep it up to date and ready to find any threats to your data and information. Configure it properly and keep the database updated and it will serve to keep your information protected from any viruses.

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